

Interactive Radio

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Broadcast radio is perceived by most artists largely as a means by which performance, pre-recorded or otherwise, reaches a mass audience. I have never found the idea of radio as the vehicle for carrying a message, musical or otherwise, particularly appealing; however, radio can become an interesting social medium when it is used in its own right and played with in much the same way that a musical instrument can be played. Cage's *Imaginary Landscape No. 4* [1] was probably responsible for the turning point in my thinking about performance using radio because it led me to question the extent to which a composer can expect to control what a listener hears. My hope is that the advent of modern digital communications technology—technology designed especially to maximise human participation—will open the way for active communal input in productions of music, creating a scratch orchestra [2] in a new communications age. Using specific examples—*Concert on Bicycles*, first done in Canberra, Australia's capital, in 1983 [3] and *Talk-Back Piano*, accomplished in 1991—I will present descriptive accounts of interactive uses of conventional broadcast radio technology.

CONCERT ON BICYCLES

Concert on Bicycles took place in Canberra in November 1983, utilizing one of the positive features of that city's planning—its numerous bicycle paths. The city is quite picturesque at that

time of year, and the weather is usually pleasant. The idea for *Concert on Bicycles* came to me while I was cycling to work at the Canberra School of Music and thinking about the problem of how to organise a presentation of electronic music without the usual anachronism of the standard concert format.

The question that bothered me most was: once all the visual stimuli enjoyed by audiences of chamber music, orchestral and operatic performances are removed, as happens in concerts of taped work, what was an audience expected to watch? While some radiophonic pieces work beautifully in a darkened room, the effect that other senses have on the way music is perceived cannot be totally ignored.

ABSTRACT

Despite prevailing program formats, broadcast radio is a medium in which an unprecedented level of communal participation can be combined with the public exposition of private sound worlds. Two of the author's recent broadcast radio events are discussed here in detail: *Concert on Bicycles* (1983–1984), a program broadcast from a community radio station to an audience of cyclists who participated by riding en masse while listening to portable radios attached to their bicycles; and *Talk-Back Piano* (1991), an interactive radio broadcast during which members of the radio audience telephoned the radio station and used vocal and other sound to influence a live-to-air improvisation taking place on a computerised player-piano called a Disklavier.

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Received 1 December 1993.

Fig. 1. *Concert on Bicycles*, map of the route taken by two groups of cyclists travelling in opposite directions around Lake Burley-Griffin. One group travelled south, then west (along the route shown by a dotted line) to Scrivener Dam before returning on the north side of the lake back to the start; the other group travelled along the same route in the opposite direction (starting with the path shown by an unbroken line). The total distance around the course was approximately 12 km.

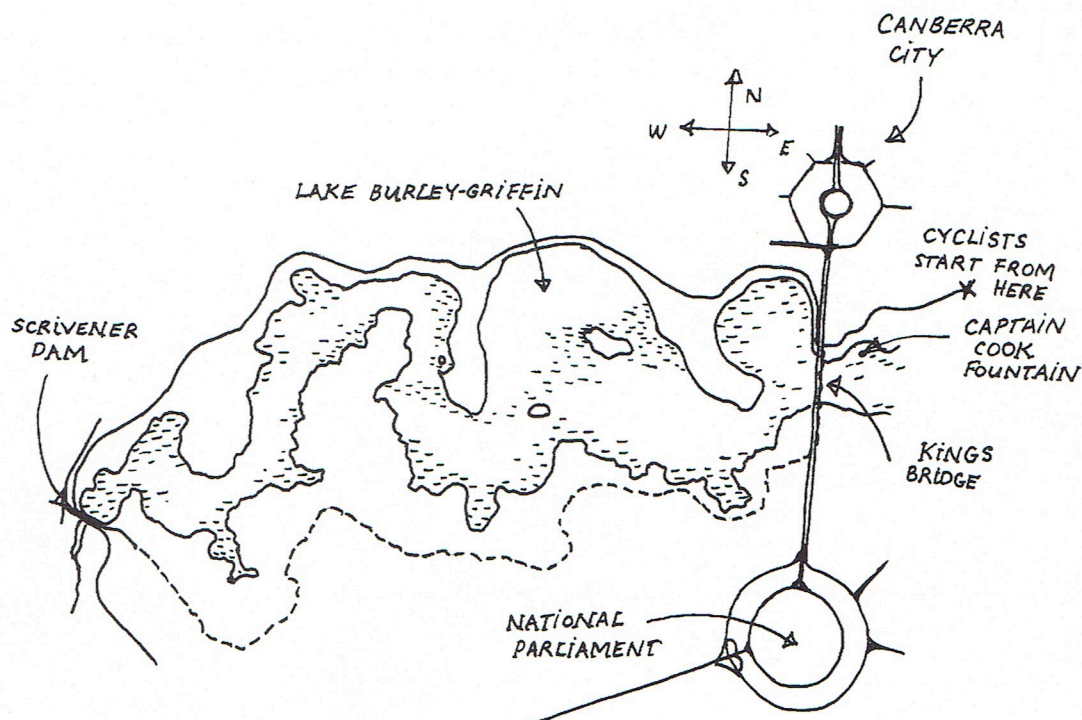




Fig. 2. *Concert on Bicycles.* Cyclists, tuned to radio station 2XX, rode around Lake Burley-Griffin in two groups, each group travelling in the opposite direction. Members of one group are seen here shortly after the start of the program. (Photo: Canberra Times)

By contrast, there is an abundance of sensory stimuli in the open air that is not easily transferable to a concert hall. So, from the outset, *Concert on Bicycles* was to be more like a picnic than a concert. Placing taped sounds in a context of sunlight, springtime, the lake and the Brindabella Ranges seen on the horizon would not only provide all the visual stimuli one could want, but would also take care of other psychological factors that work on an audience. The idea of using mobile sound sources seemed at first to be a convenient way to overcome the problem of transporting a loudspeaker and an amplification system outdoors, but I soon recognised the potential effects that one would hear once the portable sound sources began to move. Instead of planning an event in which the audience would travel to hear a concert that would begin only after they had arrived at a given location, I conceived of an event in which the audience would experience the sound while travelling.

The obvious readymade instruments necessary for an outdoor event were available in the average household: mobile transistor radio receivers—lots and lots of them. Members of the audience were allowed to bring their own amplifiers and loudspeaker systems. In cooperation with Kimmo Vennenon and 2XX-1008, a Canberra-based community radio station, I arranged for a prepared tape to be broadcast for the duration of the event. I mapped out a bicycle course to be travelled in 1 hour (Fig. 1). We decided that the event would start near Captain Cook Fountain and that the audience would be split into two groups travelling in opposite directions along paved bicycle paths around the shores of Lake Burley-Griffin. One group would travel clockwise around the lake, i.e. south over King's Bridge, west towards Scrivener Dam and then back to the

starting point; the other would travel counterclockwise on the same path. This meant that members of each group would eventually pass one another in opposite directions at roughly twice the normal speed of one cyclist. Doppler effect, the perceived pitch shifts that occur when a sound source moves relative to a listener, would thereby be increased.

I prepared a 1-hour-long tape for the event in collaboration with composition students from the Electronic Studio at the Canberra School of Music. Each composer was to produce a segment of a tape work that they believed would complement the outdoor sound environment [4] or utilise the effect of mobile sound sources [5].

Before the event, radio station 2XX ran a series of announcements. Other community arts groups, such as the organisers of Sunday in the Park and Gorman House, distributed flyers to ad-

vertise the event. The publicity gave instructions the cyclists would need to follow in order to take part and explained where and when they should assemble. The public was advised that small personal stereos with earphones were unsuitable for the event and that, instead, portable radios with external speakers were necessary to hear the combined effects of the cyclists' movement and the sound. All the publicity emphasised active communal participation—a typical caption on one of the fliers read, "So fasten your radios onto your bicycles and take your ears for a ride. They need the exercise, too!"

Sunday, 21 November, was a beautiful, clear day without a breath of wind, and approximately 130 people turned out on bicycles fitted with radios (Figs 2–6). One person even brought his dog along for the ride (see centre of Fig. 5). During the broadcast, which was in monophonic sound, effects such as flanging [6] and pseudo-stereo sound images [7] could be heard quite frequently. These effects were caused by the same sound being produced from two or more speakers moving at different speeds. Each cyclist would hear at least two versions of the sound being broadcast: one version at normal pitch, from the radio travelling on his/her bicycle; and other versions of the same sound with the pitch shifted, from just a few cents to as much as a semitone, depending on the speed of approaching bicycles. Sometimes these effects were apparent only when cyclists were a few metres apart (as close as the cyclists shown in Figs 2, 3

Fig. 3. *Concert on Bicycles.* The large radios used by some cyclists made a significant contribution to the auditory spatial effects experienced by all participants. (Photos: Canberra Times)





Fig. 4. *Concert on Bicycles*. With the lake and hills as a backdrop, one cyclist rode with his radio suspended from two flexible flag masts (commonly used by cyclists in spring to protect riders from attacks by nesting magpies). (Photo: Canberra Times)

and 5). Sound reflections off the ground, off stationary objects alongside the bicycle path and off the walls of a tunnel near Scrivener Dam accounted for some interesting effects that had not been anticipated. One of these effects was the impression of sudden aural activity produced simply by a single strong reflection off a stationary object such as a large rock or post. The cyclist would simply experience this as a moving sound whizzing past one's head as one rode by—as if a magpie swooped by.

For *Concert on Bicycles*, moving radio receivers had been used to create new aural artifacts that were not in the original prepared tape. These artifacts were a by-product of the movement of participating cyclists, and only those who had contributed to their creation by taking part on a bicycle could experience them first-hand [8].

TALK-BACK PIANO

Talk-Back Piano was conceived from my related interest in radio as an interactive medium and was realised on Australian Broadcasting Corporation (ABC) radio, 28 October 1991, as my contribution to "SoundCulture91," an international festival held in Sydney that drew together people working in various sound arts from diverse parts of the Pacific rim. In collaboration with a production team led by Andrew MacLennan and Ros Cheney of "The Listening Room," an

ABC radio program, I explored a new broadcast format developed to make it possible for many people to participate communally in playing a very large musical instrument.

Talk-Back Piano was designed to be an interactive radio broadcast involving a computer-controlled musical instrument that could be affected by audience response. The ABC program broadcast the sound from a Yamaha Disklavier [9]—a computer-operated player piano (Fig. 7)—connected to and played by a programmable device called the MIDI Tool Box (see Appendix), which converts audience response in the form of audio signals to commands that are executed on the Disklavier. Central to this production was MIDI Tool Box hardware (Fig. 8), which received voice signals over the telephone and translated them into MIDI information to drive the player-piano, effectively translating audience response into performance.

Talk-Back Piano sent sounds across vast distances to interact spontaneously with each other. The work combined broadcast radio and telephonic and digital communication to create a virtual piano that could be played by many participants over large geographical areas. The work provided a public space where interaction is possible between the private sound worlds of otherwise isolated members of the radio audience. The resultant collage of private—and, at times, very imaginative—sound worlds was made possible by a marriage of diverse

technologies. Members of the audience, without having to give up their anonymity [10], were allowed to find their own musical response from the instrument.

During the program, members of the radio audience were invited to participate in the transmission by calling one of three telephone numbers that were announced during the commentary given throughout the broadcast. These participants were able to "play" the piano by ringing up one of the numbers: the first number allowed the caller to influence the rhythm; the second number allowed the caller to influence the harmony; and the third number allowed the caller to influence the tempo at which the music was played.

The Disklavier's performance was composed of 16 polyphonic layers that were generated in real time from a table of pitches, dynamics and durations. Access to the notes in the table was controlled by a cyclic algorithm based on linear feedback shift registers.

A linear feedback shift register is a technique for generating a digital data sequence. It is a useful way to generate sequences because it can be implemented easily in binary arithmetic without using large amounts of memory and the nature of the data can be easily altered using only a single index. Linear feedback shift registers are the basis of engineering applications such as digital white noise generation, digital error-correction, data encryption and spread-spectrum modulation (used in cellular

Fig. 5. *Concert on Bicycles*. Lively binaural shifts in the broadcast sound were produced when cyclists moved in groups, a probable cause of excitement for the dog (centre), which barked playfully as other bicycles approached. (Photo: Canberra Times)



telephones). My development of linear shift register into one suitable for musical variation is described elsewhere [11]. For *Talk-Back Piano*, a further development of this algorithm uses data sequences to address look-up tables, but a full description is beyond the scope of this article.

Throughout *Talk-Back Piano*, the Disklavier played sporadically for the duration of the event—even when there were no callers. When a caller rang in to participate, the caller's sound caused the pitches, dynamics and durations to change. For example, when someone called to affect rhythm, the caller's signal was used to shift accentuation from some notes to others; when someone called to affect harmony, harmonic filters and a process used to control the tessitura (region of the keyboard) in which notes were played were shifted by the caller's signal, causing systematic changes to the harmony. When someone else called on a different line to affect the tempo, the rate at which the MIDI Tool Box produced MIDI was affected. The algorithm was programmed to produce some Disklavier activity at all times. Therefore, if there were no callers present to affect the pitch and tempo, the changes caused by the caller affecting harmony would still be perceivable.

Furthermore, all sound made by the participants (such as speech, singing or other means of musical performance)

Fig. 6. Concert on Bicycles. Doppler effects were pronounced in the second half of the broadcast, when cyclists had the opportunity to hear their own sound interact with the same sound coming from other cyclists passing in the other direction. (Photo: Canberra Times)



Fig. 7. Talk-Back Piano. The composer is seen here using a modified telephone connected to the MIDI Tool Box of the Disklavier used in *Talk-Back Piano*. (Photo: Jon Drummond)



was heard in the transmission of the radio program, along with the live piano sounds and program commentary, as an audible part of the live broadcast. Throughout the broadcast, which was designed to last for at least one hour, a program compère facilitated the interaction between participants and the musical system.

The nature of the algorithm made it impossible for one caller's gestures to dominate the instrument in a completely deterministic fashion. The system in no way pretended that people would be able to adopt—and perpetuate—the role of a soloist. Instead, callers had to intuitively learn—by trial and error—how to use the system, which is essentially an instrument meant for group performance.

Callers learned to play the Disklavier throughout the course of the broadcast by making a sound—produced with their voices or by any other means—into their telephone mouthpieces. They could sing, recite a poem, play a musical instrument, make sounds that seldom surface in other public music forums and, above all, listen to what the piano was doing in response. By listening during the moments when they were not making sounds, participants were aware of the virtual space isolating them from the piano's response to their sounds. Callers were briefed by the telephone switchboard operator before being placed in a queue to go on the air. The compère invited each caller, one at a time, to play on the air, introducing them by their given names and the localities from which they were calling.

We had planned to introduce two, and then three, callers together at a later point in the program to create sound textures with several independently controlled strands, each affecting in different ways the piano sound produced. At the climax of the program,

shortly before its conclusion, the piano was to be broadcast unaccompanied by any sound stimuli from the participants. We intended the piano to play solo for a period of several minutes and then fade out under production announcements and acknowledgments of contributors and sponsors.

The compère, Bruce Keller, provided commentary modeled after that of a talk-back radio program host. While he may have been tempted to do a send-up of some of the more objectionable media personalities, Keller instead provided a commentary that avoided both over-seriousness and unnatural humor. Keller's commentary was intended to be dispensable at the instant participants began to interact with the piano, and like a facilitator, he delivered in a tone of voice that was intended to arouse both audience curiosity and active contribution [12]. Any sense of contributions having failed was never conveyed to participants; rather, they were given a sense of a private sound world that everyone has a right to explore—not only the musician sitting in front of a large audience.

The production team also agreed to make arrangements, without publicising the fact, to deal with complaint calls by diverting them to toll-free numbers. We decided that any ardent devotees of traditional piano music affronted by the informal program content of *Talk-Back Piano* should be allowed to complain, provided they were democratic enough to do so live, on the air, complete with piano accompaniment—who needs Dada when you can have democracy? The compère, while treating complainers with respect, was not to be drawn into arguments. The decision as to how long such callers should remain on the air was to be left to the discretion of the producer.

However, even the best plans sometimes overlook the unexpected. After

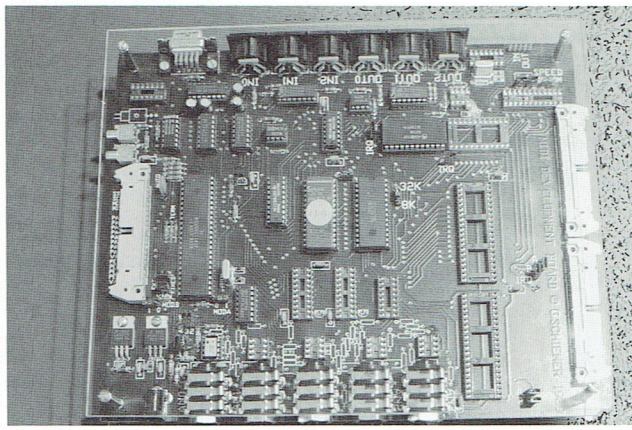


Fig. 8. *Talk-Back Piano*. The MIDI Tool Box, purpose-built hardware used to connect the audience sound to the Disklavier. The interactive algorithm resides in the ROM (a small component near centre, covered by a yellow spot). (Photo: Jon Drummond)

about 30 minutes, there was a system failure and the piano stopped in the middle of a call. Quickly, production was switched to a standby program on the music of Conlon Nancarrow and his player-piano studies [13]. The ABC switchboard was jammed for the remaining 20 minutes of the program by callers who had rung from all over the Eastern time zone and were waiting to take part, hoping that the problem would be rectified before the program was due to finish. No one knew what went wrong.

What we did know was that the participants who had gone on the air had produced unexpected responses, revealing sound worlds that were private and intimate, and had acted totally without inhibition in the public domain. The range of responses was diverse. Some were particularly memorable, such as one caller from Erskineville (a Sydney suburb) who read from Patrick White's *The Solid Mandala* [14], another from Kangaroo Valley (on the NSW south coast) who played a single glissando using a signal generator, another from the outskirts of Brisbane (in south Queensland) who played wine goblets and other assorted living-room ornaments, another from North Sydney who played a didgeridoo, another ringing from Townsville (far north Queensland) to sing, and a caller from a small town in Victoria who performed howling sounds made into the belly of a guitar. The result was not merely the creation of an extended living room, but the broadcasting of all manner of closet performance imaginable, uncensored by the tyrannies of musical convention. The audience had not only assumed a new importance; it had become interesting.

It was as if the piano's hammers, strings and frame had become an extension of each caller's vocal resonance. Radio had become a new instrument, spread over a huge area and used like a

large, collective vocal tract. The silences and the strangeness of the sounds gave the sense that the isolation and remoteness produced by immense distances had been bridged, uncensored by any notions of what was appropriate for a concert hall. Even as the system failed on the air, the feeling that overcame me while listening in the studio was one of optimism for a new medium no longer confined to one-way forms of communi-

cation. Here, perhaps, was a medium in which listeners would no longer be relegated to the role of a clientele willing to have its musical taste directed and controlled [15].

TECHNICAL APPENDIX: THE MIDI TOOL BOX

Central to the *Talk-Back Piano* production was a piece of computer hardware that I designed and built in 1989 and that I call the MIDI Tool Box (MTB) [16]. For the MIDI Tool Box, I also devised a computer program capable of improvising up to 16 polyphonically independent musical lines or parts on the player-piano. Audio signals fed into the Tool Box from telephone lines from a switchboard in the broadcast studio are converted into digital control information, allowing audience response to influence the selection of musical detail produced by the algorithm and thus alter the musical character of the improvisation that is played on the Disklavier in real time. This module contains all rules for impro-

Table 1. Applications of the MIDI Tool Box developed by various musicians and sound artists.

Application	Compositions & Installations	Composer/Researcher
MIDI Theremin	<i>Spirals</i> <i>Caged Piano</i>	Jon Drummond Greg Schiemer
MIDI-Controlled Equal-Powered Quad Panner	<i>Spirals</i>	Jon Drummond
Audio-Controlled Disklavier Interface	<i>Talk-Back Piano</i> <i>Voltage-Control Piano Studies 1—4</i>	Greg Schiemer Greg Schiemer
MIDI-Controlled Wine Glasses	<i>Glasses</i>	Stephen Parker
Improvising Machines	<i>Roulette Rhapsody</i> <i>Words, Meaning, Sounds</i> <i>Harmonicscape</i> <i>M3</i> <i>POLY_CG6</i> <i>Polyphonic Variations</i> <i>Spectral Variations</i> <i>Spectral Dance</i> <i>Token Objects</i> <i>Machine Dance</i>	Jon Drummond Andrew Peachey Fredrick Song Matthew Hoare Carlo Giacco Greg Schiemer Greg Schiemer Greg Schiemer Greg Schiemer Greg Schiemer
Sound Automata	<i>Orbits</i>	Densil Cabrera
Intelligent MIDI Merge Unit		John Tassopoulos
High-speed MIDI Local Area Network Interface		Greg Schiemer and Danny Morris

visations played by the *Talk-Back Piano* as well as the audience interface software. This program is located in a single on-board read-only memory (ROM) (the component with a small circular dot located near the centre of Fig. 8).

I designed the MIDI Tool Box to allow people working with computers and sound to design applications beyond the scope of commercial MIDI systems. It contains the necessary hardware and software tools to process audible information in a variety of media formats and is useful as a development platform for new musical applications, allowing them to be developed and tested with minimal resources. Applications can boot automatically on power-up, making the MTB useful to develop music machines that do not use a disk-operating system or console terminal for input/output (I/O) control.

The MTB's design is based on a Motorola 68HC11, a microcontroller chip whose instruction set is similar to the 6800 family [17]. Additional hardware on the MTB, combined with the many enhanced I/O peripherals and memory features of the microcontroller itself, make it highly suitable for translating between a variety of signal formats and MIDI. During program development, a host machine is used both to assemble source programs and as a terminal emulator [18] for the MTB, whose software-monitor ROM allows user-written applications to be loaded, run and debugged. The monitor ROM also contains some commonly used MIDI utilities that can be called from user programs. The host machine can be an IBM-PC, Atari, Amiga or Macintosh.

The design philosophy of the MTB facilitates low-level programming for composer(s)/performer(s) who design and construct purpose-built instruments. Table 1 shows typical applications developed to date by various musicians and sound artists. More detail on individual pieces can be found elsewhere [19].

Acknowledgments

I thank Kimmo Vennenon, of 2XX Canberra Community Radio, announcer for *Concert on Bicycles*; anonymous students and tape contributors from the Canberra School of Music, 1983; Andrew MacLennan, Ros Cheney and Tony McGregor of ABC Radio, producers of *Talk-Back Piano*; Bruce Keller, dramaturge and program compère; June Cowl, telephone receptionist during the production; Richard Dent, of ABC Radio, for technical support in phone hook-up; Phillip Uhlmann, of ABC Radio, for studio engineering; John Jacobs, for technical support prior to ABC production; and Damon McMahon, sales representative for Yamaha NSW, who made possible the loan of a Disklavier at no cost to the production team.

I would also like to thank John Duval, technical officer, School of Electronic Engineering, South Australian Institute of Technology, for assistance to me during 1989 with writing version 1.0 of the Monitor ROM, and for assistance during etching of the first hardware prototype of the MIDI Tool Box; and Jon Drummond, composition student, Conservatorium of Music, for methodically debugging the first HC11 timer programs used in this project in 1990.

References and Notes

1. J. Cage, *Imaginary Landscape #4* for 12 radios and 24 performers, Peters Edition No. 6718 (Henmar, NY: ca. 1960).
2. The Scratch Orchestra was founded by English composer Cornelius Cardew in 1969. Formed partly as a reaction against Modernism, it was a celebration of communal music-making and made no class distinction between the dedicated music professional and the dedicated amateur musician. As defined in its draft constitution, "A Scratch Orchestra is a large number of enthusiasts pooling their resources (not primarily material resources) and assembling for action (music-making, performance, edification)." See C. Cardew, ed., *Scratch Music* (Latimer, 1972).
3. The event was repeated in 1984.
4. Electronically produced sounds were, when reproduced outdoors, considered complementary to the natural sound environment if they enhanced or increased the listeners' perception of natural sounds made in that environment. This does not include, for example, the use of a radio simply to play a familiar piece of music in an outdoor setting. While such a definition of complementary sound can only be fairly subjective—and despite the fact that it was not necessary to reach consensus on such a definition—there was nevertheless considerable agreement among participants about the complementary effect certain broadcast sounds had at various points during *Concert on Bicycles*.
5. The main effects of movement of the sound were (1) Doppler effect, an apparent change of pitch due to the movement of a sound source relative to the listener; (2) chorusing, in which beats are produced by different versions of the sound—the original source and its reflection, in which the pitch is shifted a few cents away from the original; and (3) binaural phase shifts, in which both ears perceive that the position of sound is changing relative to both ears. These effects are most evident when pitches are long sustained. Tape pieces that tended to have long sustains or pitched material—as opposed to unpitched material—were placed later in the program so as to maximise the effect of the bicycles moving past one another in opposite directions.
6. Flanging is an aural effect that occurs when a listener hears two images of a sound that are not exactly in tune with each other and therefore produce audible beats. Flanging is typically accomplished through tape manipulation or non-identical digital delays. In *Concert on Bicycles*, flanging was accomplished by having multiple sound sources moving at different speeds.
7. Pseudo-stereo sound is a term that describes a sound image whose source location can be identified by the phase difference between the sound in each ear. The effect is similar to stereo, except that, in the case of pseudo-stereo sound, the phase difference is produced as the result of delays between multiple—but otherwise identical—sound sources that move relative to one another. Although these sound sources in *Concert on Bicycles* were essentially all broadcast in mono, the sound heard by the participants had some of the spatial characteristics of a true projected stereo image. The binaural artifacts were perceivable only because there were multiple receivers moving in close proximity to one another.
8. Unfortunately, one person to whom I am particularly indebted—2XX's program presenter, Kimmo

Vennenon—could only hear the sound as it was transmitted from the radio station and therefore was unable to perceive it as the audience did while actively listening during the ride around the lake.

9. A Disklavier is a hybrid of nineteenth-century piano technology and a microcomputer interface.

10. I am not referring to total anonymity here, but anonymity only in a visual sense. Participants were introduced by their given names and the geographical locations from which they were phoning. The point being made is that participants probably felt uninhibited simply because they did not feel the eyes of an audience.

11. See G. Schiemer, "Monophonic Variations: Spontaneous Vs Premeditated Composition," *NMA 6* (1988).

12. While the compère might initially be regarded as a means of directing this process—perhaps even an affirmation of traditional radio programming—it must be emphasised that, before the event, little was known about the readiness of a radio audience to participate in such events. Some form of mediation was considered necessary to ensure continuity in the broadcast.

13. Conlon Nancarrow, *Complete Studies for Player Piano* (Los Angeles, CA: 1750 Arch Records S-1786, 1981).

14. There are no definite plans to repeat *Talk-Back Piano*, but the script has been submitted to a number of radio stations both here and overseas.

15. Patrick White, *The Solid Mandala* (London: Penguin, 1969).

16. The author's development of the algorithm is described in G. Schiemer, "Monophonic Variations: Spontaneous vs. Premeditated Composition," in A. Riddell, ed., *NMA6*, (Melbourne: NMA, 1988).

17. G. Schiemer, "MIDI Tool Box: Tools for Interactive Performance," *Chroma, Journal of Australian Computer Music Association 1*, No. 15 (1994) pp. 5–10; and G. Schiemer, "MIDI Tool Box," *Sounds Australian Update*, No. 74 (March 1994) pp. 8–9.

18. *Microprocessor, Microcontroller and Peripheral Data*, Vol. 2 (Motorola, 1988) section 3, pp. 1437–1652; also *IM68HC11 Reference Manual* (Motorola, Prentice Hall, 1989).

19. A terminal emulator is a software tool that allows a communications port on a personal computer to transfer data, character by character, to or from an external computer system. In effect, the process of terminal emulation makes both typewriter keyboard and video display unit of the personal computer behave as though they were an I/O terminal on the external computer system, thereby making it possible to design computers without duplicating I/O hardware.

20. See Schiemer [16] "MIDI Tool Box: Tools for Interactive Performance" under the subheading "Profile of Work Created by Users of the MIDI Tool Box."

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